

PORTFOLIO DEVELOPMENT REPORT

Kaidampally Sai Kumar | Professional Portfolio Build Log

This report outlines the step-by-step development process, design decisions, core features, and packages used to build the professional personal portfolio website for Kaidampally Sai Kumar.

1. Step-by-Step Development Log

Phase 1: Project Initialization & Scaffold

Initialized the workspace and scaffolded the project structure using Vite's React template. Cleared boilerplate CSS and assets to ensure a clean slate.

Phase 2: Core Dependencies Configuration

Installed react-router-dom to set up the routing architecture. Configured devDependencies like gh-pages to support one-click GitHub Pages deployments.

Phase 3: Design System Variable Mapping

Established index.css as the core design token hub. Programmed light and dark mode CSS variables (colors, border radiuses, custom scrollbars, keyframe animations) to guide consistency.

Phase 4: Component & Page Construction

Created header shells (Navbar, Footer) and standalone page views (Home, About, Skills, Education, Projects, Experience, Certifications, Contact) using clean inline SVGs.

Phase 5: Single-Page Scroll Conversion

Replaced multi-page route switching in App.jsx with a sequential scrolling grid wrapper. Set HTML scroll-padding-top to 80px to prevent sticky navbar offset overlaps.

Phase 6: Navigation Programming & ScrollSpy

Refactored Navbar.jsx to execute smooth anchor scrolls. Built a React scroll listener (ScrollSpy) that calculates window.scrollY against element heights to highlight the active section.

Phase 7: Tailwind CSS v4.0 Migration

Integrated Tailwind CSS v4.0 and configured the @tailwindcss/vite plugin. Deleted local page CSS files, refactoring classes to standard Tailwind utility variables.

Phase 8: AJAX FormSubmit Email Integration

Updated Contact.jsx form submit handler to post data to FormSubmit.co. This enables background email routing to saikumarkaidampally@gmail.com without browser page redirects.

2. Key System Features

- **Single-Page Smooth Scroll:** No-refresh navigation. Clicking items in the navbar or CTA buttons glides the page smoothly to the correct section.
- **ScrollSpy active tab tracking:** Monitors scroll offsets, automatically highlighting the nav item for the section visible on the viewport.
- **Responsive Mobile UI Layout:** Breakpoints collapse grid structures to single columns and switch the navigation bar into a responsive side drawer.
- **Light & Dark Mode Switcher:** Modifies attributes on the document element, triggering dynamic color updates defined in our tailwind themes.
- **FormSubmit.co API Integration:** Sends direct email entries silently in the background, showing state banners for loading, success, and errors.
- **Downloadable Resume PDF:** Hosted locally in the public folder, enabling the download resume buttons to fetch a valid PDF file.

3. Package System Inventory & Purposes

react & react-dom (v19.2.6) [Dependency]

Provides the foundation library core, Virtual DOM reconciliation, and component state hooks.

react-router-dom (v7.18.0) [Dependency]

Provides the Router shell context wrapper, supporting URL structure constraints.

tailwindcss & @tailwindcss/vite (v4.3.1) [DevDependency]

Injects utility-first classes and compiles them directly into Vite's asset output.

gh-pages (v6.1.1) [DevDependency]

Automates bundling index assets and publishing the build folder to the gh-pages branch on GitHub.

vite (v8.0.12) [DevDependency]

Acts as the dev server builder and compiles the production bundle.

pdfkit (v0.16.0) [DevDependency]

Utility package used to compile and export this development documentation PDF.